



[csrc-sdsu/mole] MOLE v1.2.0 (Discussion #362)

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Fri, May 8, 2026 at 9:23 AM

Reply-To: csrc-sdsu/mole <reply+BRYH6DN3RTWO6FKT4QOJB5DKO5MBPEVBMXHABGH2RE@reply.github.com>

To: csrc-sdsu/mole <mole@noreply.github.com>

Cc: Tony Drummond <ldrummondlewis@sdsu.edu>, Mention <mention@noreply.github.com>

NEW in Mole v1.2.0

- [Julia Implementation](#)
 - Core mimetic implementation of [div](#), [grad](#), [lap](#) with an included example [elliptic1d.jl](#)
- Documentation improvements
- More robust boundary conditions in [addScalarBC](#)
- MOLE OSE [Governance](#) and [Organization](#) documentation
- Additional examples in both [/examples/matlab_octave](#) and [examples/cpp](#)

Be sure to visit <https://www.mole-ose.org/> for more information about the library and upcoming events.

What's Changed

- Octave compatibility update by [@jbrzensk](#) in [#9](#)
- Folder reformatting by [@jbrzensk](#) in [#10](#)
- Added Doxygen structure for C++ files by [@jbrzensk](#) in [#11](#)
- File tree restructure by [@prtkc](#) in [#17](#)
- Isolated Doxy and Sphinx output dirs + .gitignore by [@prtkc](#) in [#29](#)
- Adding convection diffusion example in C++ by [@SavannaSpayd](#) in [#36](#)
- PR For issue 22 - writing RK2 in cpp by [@Amey-kulkarni19](#) in [#27](#)
- merged parabolic1d.cpp, parabolic1d.m, and makefile to reflect the new directory structure that is in the master branch. by [@aneeshs1729](#) in [#37](#)
- PR for Issue 13 (adding licensing file information) by [@aneeshs1729](#) in [#33](#)
- Have the handling of the ARMA variable consistent by [@joehellmers](#) in [#39](#)
- Fix 'deprecated: use braced initialiser list instead' warning message by [@joehellmers](#) in [#40](#)
- Cmakefiles V2.0 by [@JananiPSrinivasan](#) in [#42](#)
- Revert "Cmakefiles V2.0" by [@JananiPSrinivasan](#) in [#43](#)
- Add executables build to .gitignore by [@joehellmers](#) in [#41](#)
- Sphinx setup initialization by [@prtkc](#) in [#38](#)
- Cmakefiles V2.0 by [@JananiPSrinivasan](#) in [#46](#)
- Fix parabolic1D for Octave compatability by [@jbrzensk](#) in [#45](#)
- Readme changes by [@prtkc](#) in [#51](#)
- Fixing error in RK2.cpp with data.txt not being created by [@edfink234](#) in [#47](#)
- ADD_Example_Schrodinger2D.cpp by [@arrowguy234](#) in [#55](#)
- Fixed missing #include, and added doxygen content by [@jbrzensk](#) in [#65](#)
- Latex pdf addition by [@prtkc](#) in [#58](#)
- Google tests and Updated Github Actions by [@JananiPSrinivasan](#) in [#71](#)
- Makefile deletion by [@JananiPSrinivasan](#) in [#79](#)
- updates boundary info into discrete analog system by [@mdumett](#) in [#66](#)
- License info modifications. by [@prtkc](#) in [#86](#)
- convection_diffusion3D | Name fix by [@prtkc](#) in [#87](#)
- Update readme.md by [@JananiPSrinivasan](#) in [#91](#)
- Fixed openBlas issue in CmakeLists by [@JananiPSrinivasan](#) in [#90](#)
- MacOS + SuperLU build fix by [@prtkc](#) in [#92](#)
- .gitignore changes corresponding to new cMake system by [@prtkc](#) in [#82](#)
- Doc builder make instructions. by [@prtkc](#) in [#93](#)
- removed finite by [@prtkc](#) in [#95](#)
- Unit_tests for MATLAB by [@JananiPSrinivasan](#) in [#96](#)
- Update CONTRIBUTING.md by [@jbrzensk](#) in [#102](#)

- [Doc] Improved landing page by @prtkc in #97
- Minor content fix by @prtkc in #106
- Patch-Fix by @JananiPSrinivasan in #109
- Add elliptic3D.cpp by @katayoonk in #61
- adding hyperbolic1D_upwind.cpp by @edfink234 in #63
- Lock exchange cpp implementation by @katayoonk in #100
- Create PULL_REQUEST_TEMPLATE.md by @jbrzensk in #101
- MD Example files + SVG Parser by @prtkc in #107
- PR for Poisson2D.cpp by @arrowguy234 in #113
- MATLAB/Octave documentation by @prtkc in #110
- Add Periodic Boundary Conditions, operators, and examples (Add BCs3 fixed) by @jbrzensk in #117
- Adding pure periodic boundary conditions and also a generic boundary condition by @mdumett in #99
- Burgers1D example in C++ by @arrowguy234 in #73
- added .nojekyll by @prtkc in #118
- Update README.me docs badge URL by @valeriabarra in #120
- README: Update docs badge to point to latest version by @valeriabarra in #123
- Add 1D and 2D Wave Equation Examples using Mimetic Methods by @Arshiallaty in #57
- badge-fix by @JananiPSrinivasan in #131
- IntelLLVM support extended by @JananiPSrinivasan in #133
- Combined builds and edited README for macOS and Windows gnuplots by @jbrzensk in #127
- 2D Parabolic by @onzerem in #124
- Readme links fix by @prtkc in #136
- Check for Fortran compiler added by @JananiPSrinivasan in #138
- Updated theme changes for the documentation by @prtkc in #134
- Example new structure by @prtkc in #121
- force link of installed gfortran by @jbrzensk in #145
- Terzaghi-1D by @JananiPSrinivasan in #146
- Doc/elliptic1 d by @jbrzensk in #142
- Renamed examples/matlab/minimal_poisson2D.m by @valeriabarra in #144
- Updated doc GHAs by @prtkc in #147
- Badge fix by @JananiPSrinivasan in #152
- Sphinx Document created for terzaghi1D by @JananiPSrinivasan in #153
- Readme.md updated for issue #156 by @JananiPSrinivasan in #160
- Updated the contributing guide by @prtkc in #151
- Add AUTHORS file by @valeriabarra in #165
- Bump pillow from 11.2.1 to 11.3.0 in /doc/sphinx by @dependabot[bot] in #164
- Revert PR #165 to allow more time to Review by @valeriabarra in #168
- rename: matlab -> matlab/octave + src changes by @prtkc in #155
- Fix security vulnerability by upgrading requests to >=2.32.4 by @JananiPSrinivasan in #169
- Prepare for master → main branch rename by @prtkc in #171
- Merge pull request #171 from csrc-sdsu/main by @jbrzensk in #173
- fixed matlab -> matlab_octave in links by @jbrzensk in #176
- Add AUTHORS file - again by @valeriabarra in #170
- README: Update Documentation badge by @valeriabarra in #181
- MOLE v1.1.0 by @prtkc in #182
- Badge fix by @prtkc in #185
- Update authors of User Manual in doc configuration file by @valeriabarra in #187
- Add Zenodo citing info by @valeriabarra in #189
- Improvements to Contributing Guide by @valeriabarra in #192
- Fix build errors by adding check for cmake, and install if not found. by @jbrzensk in #197
- Add P and Q weights to a file for use by C++ and Fortran by @joehellmersNOAA in #191
- fixed octave incompatibility with test_curl.m by @jbrzensk in #199
- NEW Governance Document Feature/webpage by @Tony-Drummond in #196
- Fixing missing figures by @Tony-Drummond in #206
- Initial Publications Page under Community by @Tony-Drummond in #209
- Add automatic GitHub stat tracking by @jbrzensk in #212
- Integration1D Documentation added by @jbrzensk in #218
- docs: standardize figure sizing via MyST {width=80%}; preserve Sphinx... by @prtkc in #215
- Rename doc/sphinx/source/api/matlab -> matlab_octave by @valeriabarra in #220
- chore:remove precalculated weights by @joehellmersNOAA in #226
- added two way wave equation comparison by @jbrzensk in #193
- Docs/fix ose org images 222 by @prtkc in #227
- Added cpp example: elliptic1DHomogeneousDirichlet.cpp by @Shane-Stevenson in #232
- specify eigen3.3.7 for mac osx workflow by @jbrzensk in #240
- add example elliptic1DLeftDirichletRightNeumann.cpp by @Shane-Stevenson in #243

- Bpw/sturm lcpp by [@bewagner1](#) in [#235](#)
- Added backward euler C++ example and documentation by [@kle96](#) in [#236](#)
- Add example elliptic1DNonHomogeneousDirichlet.cpp by [@Shane-Stevenson](#) in [#242](#)
- Add example elliptic1DLeftNeumannRightNeumann.cpp by [@Shane-Stevenson](#) in [#244](#)
- Fixing plot title error by [@Shane-Stevenson](#) in [#241](#)
- Rename elliptic1DLeftDirichletRightNeumann.m/cpp -> elliptic1DLeftneumannRightDirichlet.m/cpp by [@jbrzensk](#) in [#246](#)
- Update SuperLU and Armadillo versions by [@joehellmersNOAA](#) in [#254](#)
- error check for system gnuplot call by [@jbrzensk](#) in [#259](#)
- Update SuperLU and Armadillo versions ([#254](#)) by [@joehellmersNOAA](#) in [#264](#)
- Correction to Backward Euler C++ Example and Documentation by [@kle96](#) in [#249](#)
- Fix sturmLiouvilleLegendre bug by [@jbrzensk](#) in [#265](#)
- Feature / julia implementation by [@mathgeek31415](#) in [#237](#)
- Fix image handling for RTD builds by [@prtkc](#) in [#260](#)
- Bump pillow from 11.3.0 to 12.1.1 in /doc/sphinx by [@dependabot\[bot\]](#) in [#276](#)
- Update PR Template to be more explicit about PR/Issue pair by [@valeriabarra](#) in [#283](#)
- MOLE.jl: Refactoring in Opearors and BCs Modules by [@valeriabarra](#) in [#282](#)
- Closes [#284](#) regarding adding scalar boundary conditions by [@Jiya-Rathi](#) in [#290](#)
- New PR closes [#284](#) regarding adding scalar boundary conditions by [@Jiya-Rathi](#) in [#298](#)
- julia/MOLE.jl: Add addScalarBC, relative BCs tests and elliptic1DaddScalarBC example by [@valeriabarra](#) in [#300](#)
- Renamed uppercase example files to lowercase by [@jbrzensk](#) in [#277](#)
- Fix: Replace file(GLOB) with explicit source lists in CMakeLists ([#203](#)) by [@Jiya-Rathi](#) in [#314](#)
- Remove gallery images from README and repository by [@jcorbino](#) in [#322](#)
- Implement clang-tidy action. by [@cpaolini](#) in [#316](#)
- Fixed clang-tidy errors in Schrodinger2D.cpp by [@jbrzensk](#) in [#324](#)
- C++ 8th Order Divergence by [@bewagner1](#) in [#325](#)
- Feat: Add 1D Wave convergence example (Matlab & C++) ref Mathews & Fi... by [@tin7](#) in [#270](#)
- Adding cpp&MATLAB example - 2D incompressible square cylinder flow by [@yiyuef](#) in [#274](#)
- Merging CPP_Mim_Ops_New in CPP_Mim_Ops to complete the mimetic operators as in MATLAB/octave by [@Jiya-Rathi](#) in [#338](#)
- Add mixed periodicity to MATLAB interpolators and implements interpolators in C++ by [@bewagner1](#) in [#318](#)
- Revert "Merging CPP_Mim_Ops_New in CPP_Mim_Ops to complete the mimetic operators as in MATLAB/octave" by [@Tony-Drummond](#) in [#345](#)
- Fix paths for MOLE.jl docs by [@valeriabarra](#) in [#348](#)
- Additional Unit Tests (MATLAB) by [@mathgeek31415](#) in [#333](#)
- Fix error calculation in MATLAB_Octave tutorial by [@jbrzensk](#) in [#350](#)
- Enable CTest to capture assertSuccess status by [@cpaolini](#) in [#346](#)
- Feature/cpp mim ops by [@Jiya-Rathi](#) in [#336](#)
- Fixes for Issue related to MATLAB/Octave documentation Issue [#302](#) by [@deepanshu-zade](#) in [#341](#)
- Added python script to automatically propagate the MOLE software license in root directory to all MATLAB/Octave scripts and C++ files by [@deepanshu-zade](#) in [#342](#)
- Removing out of place subdirectory by [@Tony-Drummond](#) in [#356](#)
- Adding Software Design Document to [mole-docs.readthedocs.io](#) by [@Tony-Drummond](#) in [#352](#)
- Removed commented lines inside index.html by [@Tony-Drummond](#) in [#359](#)

New Contributors

- [@prtkc](#) made their first contribution in [#17](#)
- [@SavannaSpayd](#) made their first contribution in [#36](#)
- [@Amey-kulkarni19](#) made their first contribution in [#27](#)
- [@aneeshs1729](#) made their first contribution in [#37](#)
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- [@JananiPSrinivasan](#) made their first contribution in [#42](#)
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- [@Arshiallaty](#) made their first contribution in [#57](#)
- [@onzerem](#) made their first contribution in [#124](#)
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Full Changelog: [v1.0...v1.2.0](#)

This discussion was created from the release [MOLE v1.2.0](#).